Supplementary:search strategies

Web of science Search strategy

#4 #1AND #2 AND #3
#3 TS=("Neoadjuvant chemotherapy")
#2 (((((((((TS=("Quality of Life")) OR TS=("life quality")) OR TS=(QOL)) OR TS=(HRQOL)) OR TS=(Support)) OR TS=(depress*)) OR TS=(anxiety)) OR TS=(sexual*)) OR TS=(psychological)) OR TS=(mental)
#1 (((TS=("Breast Neoplasms")) OR TS=("breast tumor")) OR TS=("breast cancer")) OR TS=("breast carcinoma")

Pubmed Search strategy

#8 #4 AND #5 AND #7
#7 #6 OR #2
#6 "psychological"[Title/Abstract] OR "mental"[Title/Abstract] OR "depress*"[Title/Abstract] OR "HRQOL"[Title/Abstract] OR "life quality"[Title/Abstract]
#5 #1OR #3
#4 "Neoadjuvant Therapy"[MeSH Terms]
#3 "breast tumor"[Title/Abstract] OR "breast cancer"[Title/Abstract] OR "breast carcinoma"[Title/Abstract]
#2 "Quality of Life"[MeSH Terms] OR "Social Support"[MeSH Terms] OR "Depression"[MeSH Terms] OR "Anxiety"[MeSH Terms] OR "Sexuality"[MeSH Terms]
#1 "Breast Neoplasms"[MeSH Terms]

Embase Search strategy

#9 #6 AND #7 AND #8
#8 'neoadjuvant chemotherapy'/exp
#7 'breast tumor'/exp
#6 #1 OR #2 OR #3 OR #4 OR #5
#5 'sexuality'/exp
#4 'depression'/exp
#3 'anxiety'/exp
#2 'quality of life'/exp
#1 'social support'/exp

Cochrane Library Search strategy

#9 #6 AND #7 AND #8
#8 #1 OR #2 OR #3 OR #4 OR #5
#7 MeSH descriptor:[Neoadjuvant Therapy] explode all trees
#6 MeSH descriptor:[Breast neoplasms] explode all trees
#5 MeSH descriptor:[Sexuality] explode all trees
#4 MeSH descriptor:[Depression] explode all trees
#3 MeSH descriptor:[Anxiety] explode all trees
#2 MeSH descriptor:[Social Support] explode all trees
#1 MeSH descriptor:[Quality of life] explode all trees

Scopus/PsyCINFO/CINAHL Search strategy

#4 #1 AND #2 AND #3

#3 "psychological" OR "mental" OR "anxiety" OR "depress*" OR "quality of life" OR "HRQOL" OR "QOL" OR "life quality" OR "sexual*" OR "support"

#2 "neoadjuvant chemotherapy"

#1 "breast cancer" OR "breast tumor" OR "breast carcinoma" OR "breast neoplasms"

Supplementary Table S1 *Quality of RCT studies as assessed by Cochrane collaboration's tool for assessing risk of bias*

Study	Random Sequence Generation (selection bias)	Allocation Concealment (selection bias)	Blinding of Participants and Personnel (performance bias)	Blinding of Outcome Assessors (detection bias)	Completeness of Reporting Outcome Data (attrition bias)	Selective Outcome Reporting (reporting bias)	other bias
Walker et al.[26]	Low risk	Low risk	Low risk	Unclear risk	Low risk	Low risk	Unclear risk
Souza et al.[27]	Low risk	Unclear risk	Unclear risk	Low risk	Low risk	Low risk	Low risk
Basen-Engquist et al.[28]	Low risk	Unclear risk	Unclear risk	Unclear risk	Low risk	Low risk	Low risk

Supplementary Table S2 *Quality of cohort studies as assessed by Newcastle Ottawa Score(NOS)*

Study	Representativeness of exposed cohort	Selection non-exposed cohort	Ascertainment of exposure	Outcome of interest	Comparability	Assessment of outcome	Length of follow up	Adequacy of follow up	Total score
Ding et al.[20]	*	-	*	-	-	*	*	*	5
Lee KM et al.[21]	*	-	*	-	-	*	*	*	5
Takada et al.[29]	*	*	*	-	**	*	*	*	8
Chellappan[22]	*	*	*	-	**	*	*	*	8
Hermelink et al.[23]	*	-	*	-	-	*	*	*	5
Rezapour et al.[24]	*	*	*	-	*	*	*	*	7
Coelho et al.[25]	*	*	*	-	-	*	*	*	6
Lee S et al.[31]	*	-	*	-	*	*	*	*	6

Supplementary Table S3 *Quality of cross-sectional study as assessed by JBI Critical Appraisal Checklist*

Items	Aprilianto et al.[30]
1. Were the criteria for inclusion in the sample clearly defined?	N
2. Were the study subjects and the setting described in detail?	Y
3. Was the exposure measured in a valid and reliable way?	Y
4. Were objective, standard criteria used for measurement of the condition?	Y
5. Were confounding factors identified?	Y
6. Were strategies to deal with confounding factors stated?	N
7. Were the outcomes measured in a valid and reliable way?	Y
8. Was appropriate statistical analysis used?	Y
Overall appraisal	Include

Y: yes. *N*: no.